

AIR POLLUTION MONITORING SYSTEM USING MQ2 GAS SENSOR

A PROJECT REPORT

Submitted by

MATHESH NISHANTH B [24MER030]

MOHAMMED SABEER S [24MER031]

PRAAVIN RAJAA G [24MER039]

PRASANNA T [24MER042]

DEVA PRIYAN R [24MEL064]

SATHISH KUMAR K [24MEL067]

DEPARTMENT OF MECHANICAL ENGINEERING



(Autonomous)

PERUNDURAI ERODE – 638 060

1. Project Overview

The project focuses on designing and simulating an air pollution monitoring system capable of detecting harmful gases present in the surrounding environment. An MQ-2 gas sensor is used to sense gases such as smoke, LPG, methane, and other combustible gases. An ESP32 microcontroller processes the sensor output and displays the detected air-quality level on an OLED/LCD display.

The system provides a simple and low-cost solution for monitoring air pollution in industrial areas, workshops, laboratories, and environmental safety applications. When the gas concentration exceeds a predefined threshold, an LED indicator can be activated to warn the user.

Main Objectives

1. Detect harmful gases and smoke using the MQ-2 sensor.
2. Continuously monitor the surrounding air quality.
3. Process sensor readings using an ESP32 microcontroller.
4. Display the gas level and pollution status on a display.
5. Provide an early warning when gas concentration becomes high.
6. Develop a low-cost and compact monitoring prototype.
7. Improve safety in industrial and environmental applications.

2. Problem Statement

Air pollution is one of the major environmental and industrial safety concerns caused by the presence of harmful gases, smoke, and other pollutants in the atmosphere. Continuous exposure to polluted air can affect human health and may lead to respiratory problems, reduced workplace safety, and environmental damage. In industrial environments, leakage of combustible gases such as LPG, methane, and other volatile gases can also create serious fire and safety hazards. However, conventional air-quality monitoring systems are often expensive, complex, and unsuitable for small-scale applications or educational prototypes.

There is therefore a need for a low-cost, compact, and easy-to-use air pollution monitoring system that can continuously detect harmful gases and provide immediate information about the surrounding air condition. The proposed system uses an MQ-2 gas sensor to detect smoke and combustible gases and an ESP32 microcontroller to process the sensor readings. The measured values are displayed through an OLED/LCD display, allowing users to monitor the air condition in real time. LED indicators provide a simple visual warning when gas concentration increases beyond a predefined threshold. The system aims to provide an affordable monitoring solution for industrial workplaces, laboratories, workshops, and environmental safety applications, helping users identify potentially unsafe air conditions at an early stage.

3. Proposed Solution

The proposed system is a low-cost and real-time Air Pollution Monitoring System designed to detect harmful gases and smoke in the surrounding environment. The system uses an MQ-2 gas sensor to sense the presence and variation of combustible gases and smoke. An ESP32 microcontroller collects the analog sensor signal, processes the readings, and determines the pollution level based on predefined threshold values.

The detected gas level is displayed on an OLED/LCD display, allowing the user to monitor the air condition continuously. Two LED indicators are used to provide visual alerts: a green LED indicates a normal air condition, while a red LED indicates an increased gas concentration or potentially unsafe condition.

The system operates continuously and provides an immediate warning when the detected gas level exceeds the selected threshold. This makes the prototype useful for demonstrating air-quality and gas-leak monitoring in industrial areas, workshops, laboratories, and environmental monitoring applications.

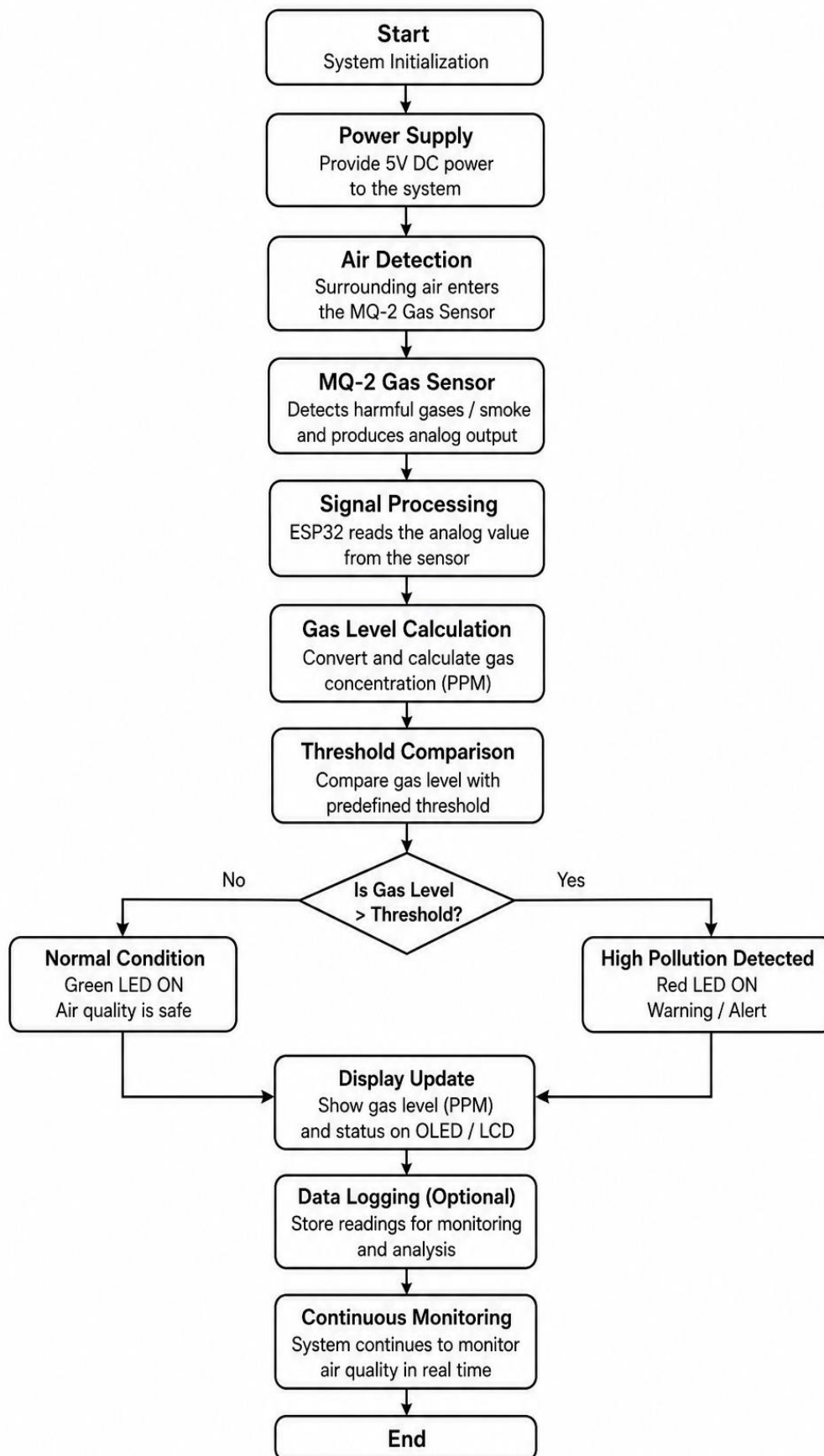
Main Features of the Proposed Solution

1. **Real-Time Gas Detection** – Continuously detects smoke and combustible gases using the MQ-2 gas sensor.
2. **ESP32-Based Processing** – The ESP32 receives sensor readings and processes them to determine the pollution condition.
3. **Air Quality Monitoring** – Monitors changes in gas concentration and provides an indication of the surrounding air condition.
4. **OLED/LCD Display** – Displays the sensor readings and pollution status clearly for easy monitoring.
5. **Visual Warning System** – LED indicators show different conditions, such as normal and high gas concentration.
6. **Industrial and Environmental Applications** – Can be used for demonstration and monitoring applications in workshops, laboratories, industrial areas, and environmental safety projects.

Overall Solution

The overall system continuously monitors air conditions and provides early warning of potentially unsafe gas levels. Its simple design, low cost, and easy implementation make it suitable for industrial areas, laboratories, workshops, and environmental safety applications.

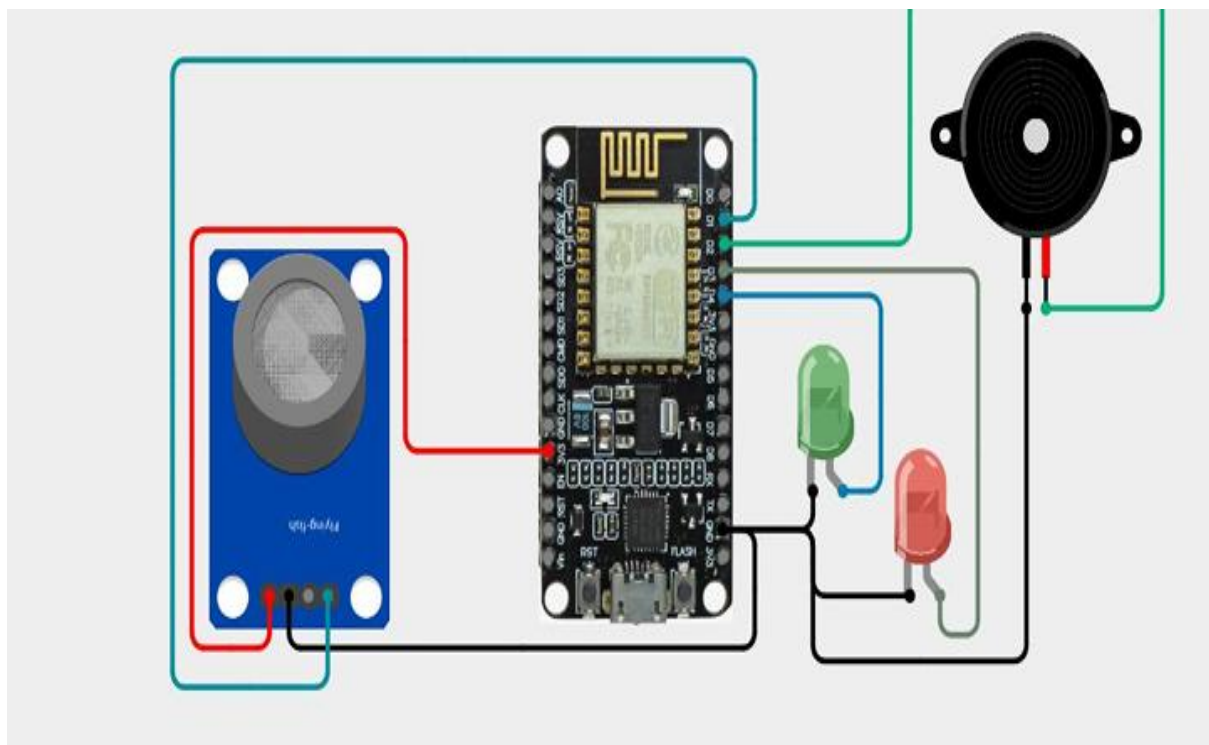
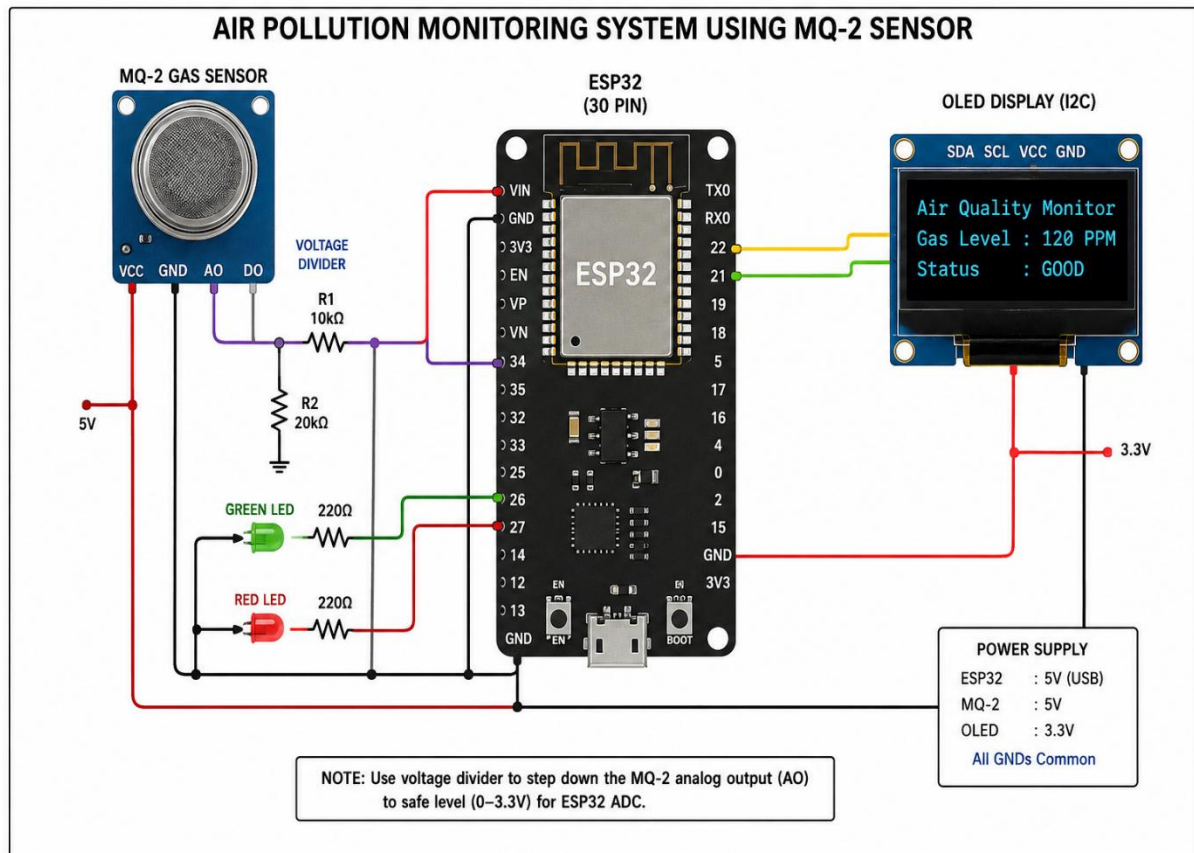
4. System Architecture



5. Circuit Table

Component	Pin	ESP32 Pin	Connection / Function
MQ-2 Gas Sensor	VCC	5V / VIN	Power supply
	GND	GND	Ground
	AO	GPIO 34	Analog gas reading
OLED Display (I ² C)	VCC	3.3V	Power supply
	GND	GND	Ground
	SDA	GPIO 21	I ² C data
	SCL	GPIO 22	I ² C clock
Green LED	Anode (+)	GPIO 25	Normal condition
	Cathode (-)	GND through 220 Ω	Ground
Red LED	Anode (+)	GPIO 26	High pollution warning
	Cathode (-)	GND through 220 Ω	Ground

6. Circuit Design



7.Algorithm

Step 1: Start the System

Initialize the ESP32 microcontroller and configure the required GPIO and ADC pins.

Step 2: Initialize Components

Initialize the MQ-2 gas sensor, OLED/LCD display, and LED indicators. Set the display to show the system startup message.

Step 3: Warm-Up the MQ-2 Sensor

Allow the MQ-2 sensor to warm up so that its sensing element becomes stable before taking measurements.

Step 4: Read Gas Sensor Data

Read the analog output from the MQ-2 sensor through the ESP32 ADC pin. The sensor output changes according to the concentration of smoke and combustible gases.

Step 5: Process the Sensor Signal

The ESP32 processes the analog reading and converts it into a suitable gas-level value for monitoring. A voltage divider is used to keep the MQ-2 analog output within the safe voltage range of the ESP32 ADC.

Step 6: Compare with Threshold

Compare the measured gas level with a predefined threshold value.

Step 7: Determine Air Condition

- If the gas level is **below the threshold**, classify the condition as **Normal/Good** and turn ON the green LED.
- If the gas level is **above the threshold**, classify the condition as **High Pollution/Warning** and turn ON the red LED.

Step 8: Display the Result

Display the measured gas level and air-quality status on the OLED/LCD display.

Step 9: Continuous Monitoring

Repeat the sensor reading, processing, comparison, and display operations continuously at regular intervals.

Step 10: End/Repeat

The system continues monitoring as long as power is supplied, providing real-time information about changes in gas concentration.

8.Coding

```
#include <Wire.h>

#include <Adafruit_GFX.h>

#include <Adafruit_SSD1306.h>


// ----- PIN DEFINITIONS -----

#define MQ2_PIN    34

#define RED_LED    26

#define GREEN_LED  27


#define OLED_SDA   21

#define OLED_SCL   22


// ----- OLED SETTINGS -----

#define SCREEN_WIDTH 128

#define SCREEN_HEIGHT 64

#define OLED_RESET  -1


Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire,
OLED_RESET);


// ----- GAS THRESHOLD -----

// Adjust this value after testing your MQ-2 sensor

int gasThreshold = 1800;


void setup()

{

  Serial.begin(115200);
```



```

// LED pins
pinMode(RED_LED, OUTPUT);
pinMode(GREEN_LED, OUTPUT);

// Start with green LED ON
digitalWrite(GREEN_LED, HIGH);
digitalWrite(RED_LED, LOW);

// Start I2C OLED
Wire.begin(OLED_SDA, OLED_SCL);

if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C))
{
    Serial.println("OLED not found!");
    while (1);
}

display.clearDisplay();
display.setTextColor(SSD1306_WHITE);

display.setTextSize(2);
display.setCursor(10, 10);
display.println("Gas");

display.setCursor(10, 35);
display.println("Monitor");

display.display();

```

```

    delay(2000);
}

void loop()
{
    // Read MQ-2 analog value
    int gasValue = analogRead(MQ2_PIN);

    Serial.print("MQ-2 Gas Value: ");
    Serial.println(gasValue);

    // ----- GAS DETECTION -----
    if (gasValue >= gasThreshold)
    {
        // Gas detected
        digitalWrite(RED_LED, HIGH);
        digitalWrite(GREEN_LED, LOW);

        display.clearDisplay();

        display.setTextSize(2);
        display.setCursor(0, 0);
        display.println("WARNING!");

        display.setTextSize(1);
        display.setCursor(0, 28);
        display.print("Gas Value: ");
        display.println(gasValue);
    }
}

```

```
display.setCursor(0, 45);  
display.println("Gas Detected");  
  
display.display();  
}  
else  
{  
    // Normal condition  
    digitalWrite(RED_LED, LOW);  
    digitalWrite(GREEN_LED, HIGH);  
  
    display.clearDisplay();  
  
    display.setTextSize(2);  
    display.setCursor(15, 0);  
    display.println("NORMAL");  
  
    display.setTextSize(1);  
    display.setCursor(0, 28);  
    display.print("Gas Value: ");  
    display.println(gasValue);  
  
    display.setCursor(0, 45);  
    display.println("Air is Safe");  
  
    display.display();  
}  
delay(2);  
}
```

9. System Working

The Air Pollution Monitoring System using MQ-2 Gas Sensor works by continuously sensing harmful gases and smoke present in the surrounding environment. The MQ-2 gas sensor detects gases such as LPG, methane, smoke, and other combustible gases and produces an analog output according to the detected gas concentration.

The analog output of the MQ-2 is connected to the ESP32 ADC pin through a voltage divider to ensure that the input voltage remains within the safe range of the ESP32. The ESP32 reads the sensor signal and processes the data to determine the relative gas level. The measured value is then compared with a predefined threshold.

When the detected gas level is below the threshold, the system considers the condition normal and turns ON the green LED. When the gas level exceeds the threshold, the system identifies a potentially high-pollution condition and turns ON the red LED as a warning.

The OLED/LCD display continuously shows the gas level and system status. The ESP32 repeats this process continuously, enabling real-time monitoring and early detection of unsafe gas conditions.

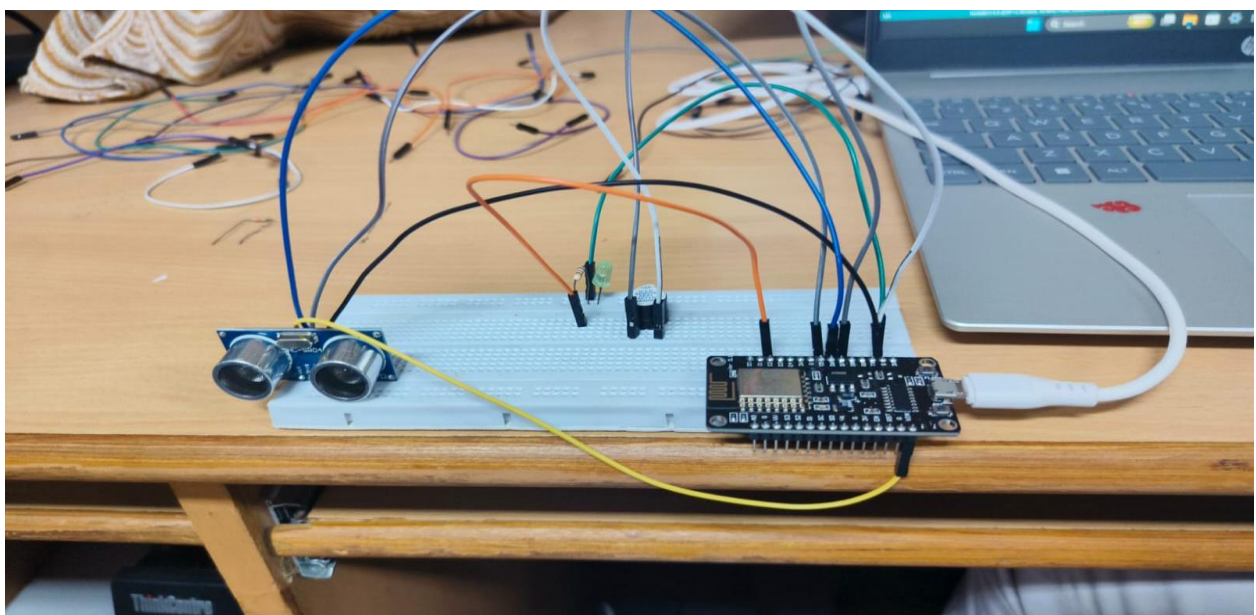
10. Bill of Materials

S. No.	Component	Specification	Quantity
1	ESP32 Development Board	ESP32, 30-pin	1
2	MQ-2 Gas Sensor	Gas/Smoke detection module	1
3	OLED Display	0.96-inch, I ² C	1
4	Green LED	5 mm	1
5	Red LED	5 mm	1
6	Resistor	220 Ω	2
7	Resistor	10 k Ω	1
8	Resistor	20 k Ω	1
9	Breadboard	Standard solderless	1
10	Jumper Wires	Male–Male / Male–Female	As required
11	USB Cable	ESP32 compatible	1

11. Prototype Implementation

The prototype of the Air Pollution Monitoring System using MQ-2 Gas Sensor is developed by assembling the required electronic components on a breadboard. The ESP32 30-pin development board is used as the main controller, while the MQ-2 sensor detects harmful gases and smoke in the surrounding environment.

- Component Assembly – Place the ESP32, MQ-2 sensor, OLED display, LEDs, resistors, and connecting wires on the breadboard.
- MQ-2 Connection – Connect the MQ-2 VCC to 5 V, GND to common ground, and its analog output through the 10 k Ω /20 k Ω voltage divider to ESP32 GPIO 34.
- OLED Connection – Connect the OLED using the I²C interface: SDA to GPIO 21, SCL to GPIO 22, VCC to 3.3 V, and GND to common ground.
- LED Connection – Connect the green LED to GPIO 25 and the red LED to GPIO 26 through 220 Ω current-limiting resistors.
- Programming – Upload the monitoring program to the ESP32 using Arduino IDE and configure the required sensor threshold.
- Sensor Warm-Up – Allow the MQ-2 sensor to warm up before taking reliable readings.
- Testing – Introduce a small amount of smoke or a suitable test gas near the sensor and observe the change in readings.
- Output Monitoring – The OLED displays the gas level and status. The green LED indicates normal conditions, while the red LED indicates a high gas-level warning.
- Continuous Operation – The ESP32 repeatedly reads the sensor, processes the data, updates the display, and controls the LEDs for continuous monitoring.



12.Result and Performance Analysis

The developed Air Pollution Monitoring System using an MQ-2 Gas Sensor and ESP32 successfully detects changes in the concentration of smoke and combustible gases in the surrounding environment. The MQ-2 sensor provides an analog signal to the ESP32, which processes the sensor reading and compares it with a predefined threshold.

The measured gas level and pollution status are displayed on the OLED/LCD display in real time. The LED indicators provide an additional warning: the green LED indicates a normal condition, while the red LED indicates that the detected gas level has exceeded the selected threshold. The prototype successfully demonstrates continuous monitoring and early warning of potentially unsafe gas conditions.

Test Condition	Sensor Response	System Status	LED Indication
Clean/Normal Air	Low reading	Normal	Green ON
Slight Smoke/Gas	Increased reading	Moderate	Green/Warning
High Smoke/Gas	High reading	High Pollution	Red ON
Gas Removed	Reading decreases	Returns toward Normal	Green ON

